ALFIDO TECH

Customer Transaction & Behavior Analysis

Customer segmentation • Purchase patterns • Retention & churn-risk insights

TRANSACTIONS	CUSTOMERS	RECORDED PURCHASE AMOUNT
250,000	49,661	681,346,299

Analysis period: 01 January 2020 to 13 September 2023

Prepared for: Alfido Tech

Dataset: ecommerce_customer_data_large.csv (provided ZIP archive)

Purpose

This report turns transaction-level data into practical customer groups and engagement priorities. It documents the cleaning approach, explains the RFM segmentation, visualizes sales and retention behavior, and proposes measurable next steps.

Important interpretation note: Amounts are shown in source units because the dataset does not state a currency. Inactivity for more than 90 days is a risk proxy, not proof of churn. Recommendations should be tested before being rolled out to all customers.

1. Executive Summary

The cleaned dataset contains **250,000 transaction rows** and **49,661 unique customers**. Recorded purchase amounts total **681,346,299**. Average recorded transaction amount is 2,725.4; average transaction rows per customer is 5.03.

RFM segmentation identifies six actionable groups. The largest is **Hibernating** (11,222 customers; 22.6%). **71.6%** of customers have had no recorded purchase for more than 90 days relative to the final dataset date. The supplied Churn label is approximately **20.0%** and is not equivalent to the inactivity proxy.

Top insights

• The largest revenue category is **Home**, contributing 25.1% of recorded purchase amount.
• The transaction series varies month to month; the highest observed month is Aug 2020 (5,886 rows).
• Customer groups differ in recency, frequency and spend, making segment-specific engagement more suitable than one campaign for everyone.
• The provided churn label has near-zero linear correlation with basic RFM features in this dataset, so it needs validation before predictive use.

2. Dataset and Data Preparation

The primary file was *ecommerce_customer_data_large.csv*. It contains 250,000 raw rows. Cleaning parsed Purchase Date, converted numeric fields, removed rows missing customer/date/amount/quantity, removed exact duplicate rows, and excluded non-positive quantities or negative total amounts. It created Order Month and Revenue per Unit, and filled missing Age from Customer Age where available.

Data quality: Returns is missing in 19.0% of raw records. Missing return values should not automatically be interpreted as zero returns. Age and Customer Age appear duplicated in the source schema. No explicit order ID is used here, so Frequency is transaction-row count, which may not always equal distinct orders.

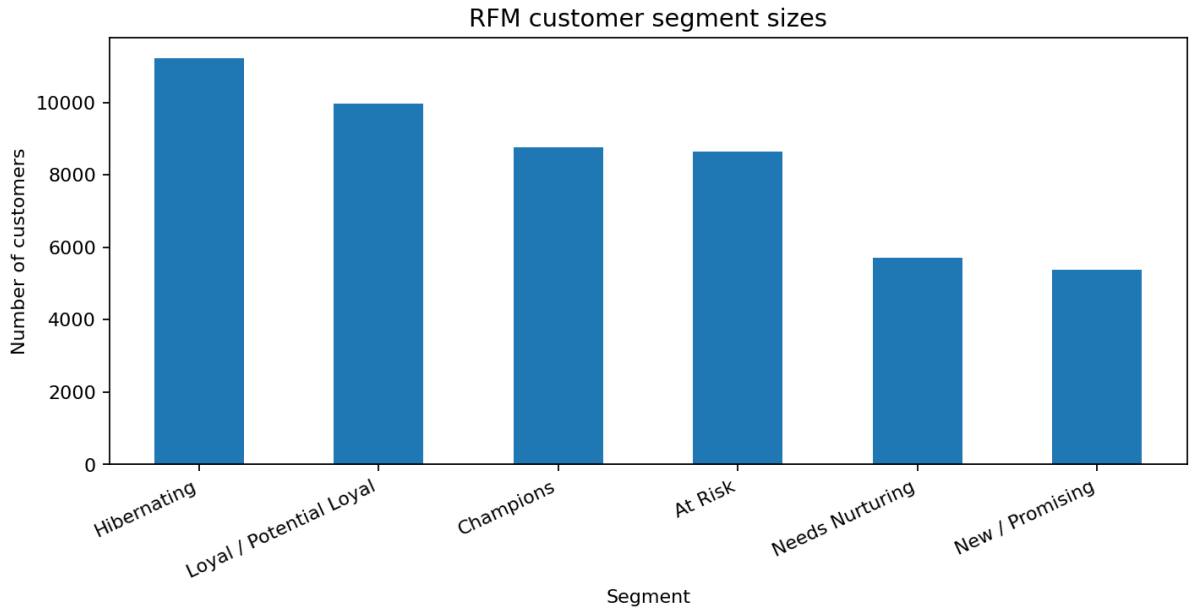
Feature definitions

• **Recency:** days between the latest recorded purchase date and the day after the dataset's last purchase date.
• **Frequency:** number of transaction rows per customer.
• **Monetary:** sum of Total Purchase Amount per customer.
• **RFM scores:** quintile scores from 1–5. Higher Recency score means a more recent purchase; higher Frequency and Monetary scores mean more activity and spend.

3. Customer Segmentation (RFM)

Segment rules are transparent heuristics for marketing prioritization, not a supervised prediction model. They combine R, F and M quintile scores into six groups.

Segment	Customers	Share	Avg recency (days)	Avg txn rows	Avg spend	Label churn
Hibernating	11,222	22.6%	564	2.8	7,623	20.0%
Loyal / Potential Loyal	9,957	20.0%	137	6.1	16,799	20.3%
Champions	8,763	17.6%	61	7.6	21,655	20.6%
At Risk	8,642	17.4%	413	5.9	16,184	19.5%
Needs Nurturing	5,704	11.5%	155	4.0	9,000	20.2%
New / Promising	5,373	10.8%	67	3.2	8,854	19.2%



Segment profiles and suggested treatment

Champions: Recent, frequent and high-spend customers. Protect with loyalty perks, early access and priority support; avoid unnecessary blanket discounts.

Loyal / Potential Loyal: Good engagement and spend, but not all scores are top-tier. Encourage habit with personalized bundles, loyalty milestones and relevant cross-sells.

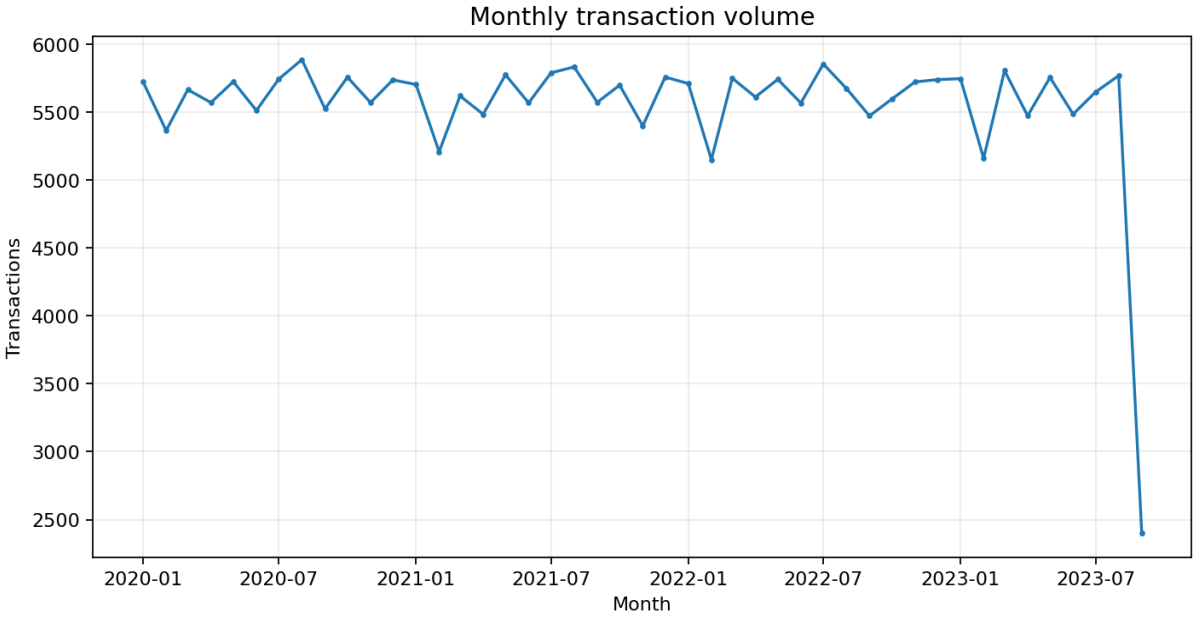
At Risk: Historically frequent customers whose latest purchase is relatively old. Prioritize win-back outreach and address likely friction before offering incentives.

Hibernating: Low frequency and long time since last purchase. Use low-cost reactivation tests; suppress expensive incentives when response is unlikely.

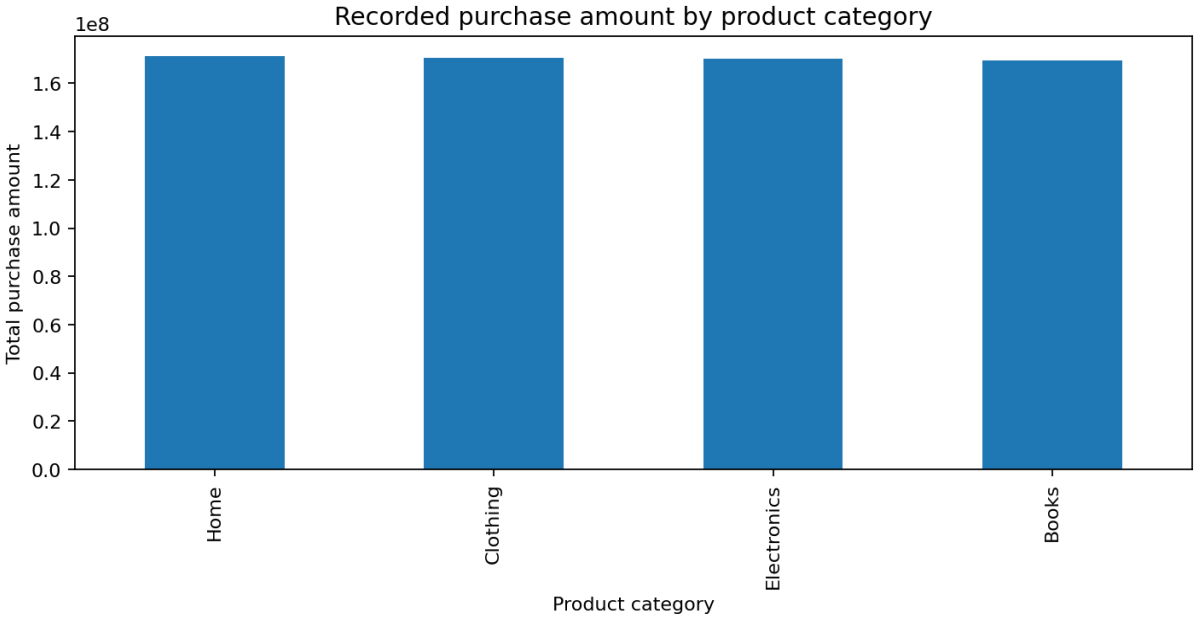
New / Promising: Recent customers with relatively low frequency. Focus on onboarding, post-purchase education and a reason to make a second purchase.

Needs Nurturing: Middle-pattern customers who do not fit the stronger groups. Use category relevance, helpful content and measured offers to move them toward repeat behavior.

4. Purchase Patterns

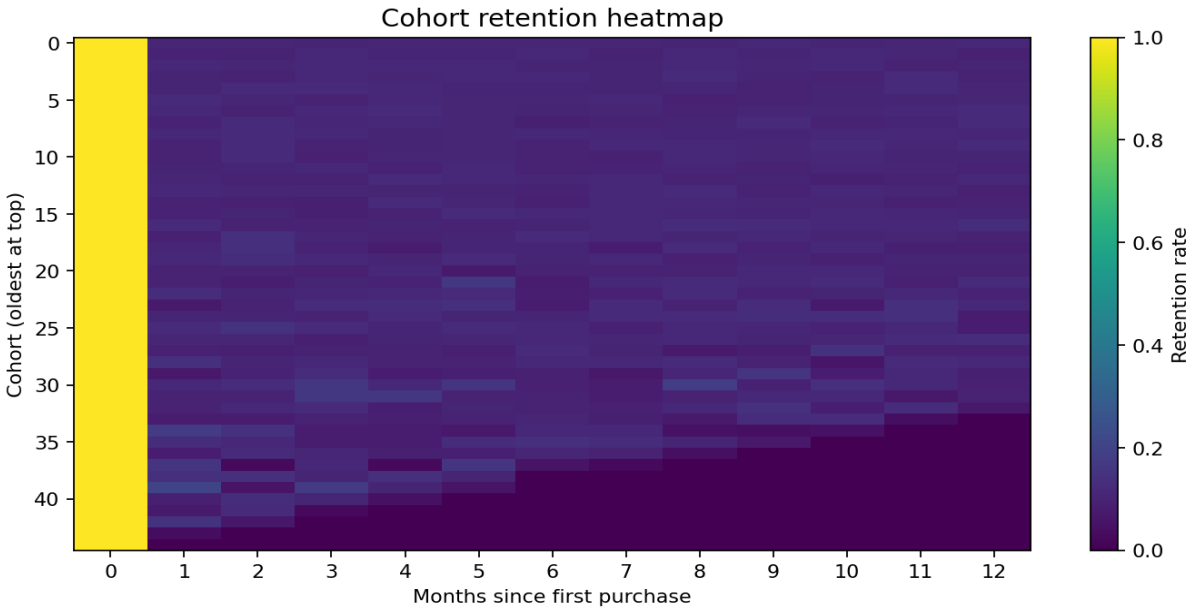


Monthly transaction volume peaked in August 2020 and was lowest in September 2023. Interpret the final month cautiously because it may represent a partial month at the dataset boundary.

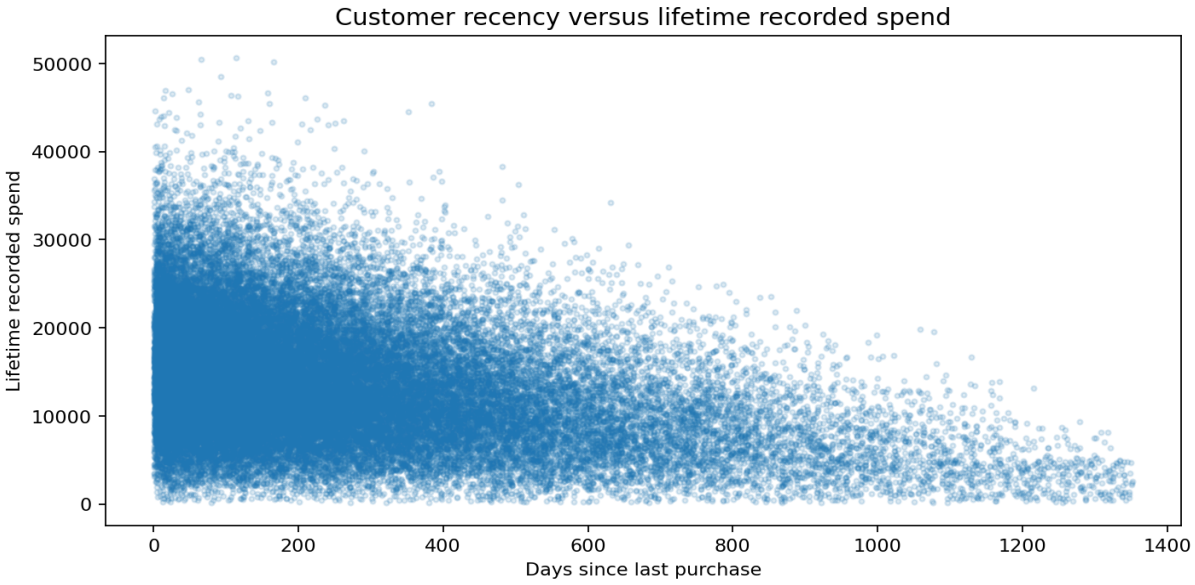


Category revenue is calculated by summing Total Purchase Amount across transaction rows. Category-level campaigns should be informed by both revenue and customer-level repeat behavior; a high revenue share alone does not prove higher margin or better retention.

5. Retention and Churn-Risk Analysis



The cohort heatmap measures the share of customers in each first-purchase-month cohort who were observed purchasing again in each subsequent month. Blank/right-edge periods reflect months not yet observed for newer cohorts; they are not zero retention.



Risk signals: 71.6% of customers have Recency > 90 days. The provided customer-level Churn label rate is 20.0%. Correlation of that label with Recency is 0.001, Frequency is 0.003, and Monetary is 0.004. These near-zero relationships mean the label's definition, timing, or construction should be checked before it is used as a predictive target.

6. Five Actionable Recommendations

1. Launch a measured win-back program
Create 30-, 60- and 90-day inactivity journeys. Tailor message and products to past category behavior. Test incentive vs. no-incentive groups and measure incremental return rate, revenue and margin.

2. Protect Champions and loyal customers
Offer early access, loyalty milestones, personalized service and fast issue resolution. Monitor changes in recency and purchase frequency to detect disengagement early.

3. Build the second-purchase journey
For new customers, send useful setup/use tips and complementary recommendations after purchase. Track second-purchase conversion and time-to-next-purchase by category.

4. Make retention a monthly operating metric
Publish a dashboard for cohort retention, repeat purchase rate, 30/60/90-day inactivity, revenue by RFM segment, and campaign lift. Define churn using the actual buying cycle and a documented time window.

5. Improve data quality and experiment discipline
Investigate missing Returns values, document the duplicate Age fields, and add a stable order ID and channel/campaign exposure where possible. Use randomized holdouts and compare outcomes by segment, category and payment method.

7. Measurement Plan

Goal	Primary KPI	Suggested review
Reactivation	Share of targeted inactive customers who purchase again; incremental revenue	Weekly during test
Loyalty	Repeat purchase rate and revenue per Champion	Monthly
Second purchase	Second-purchase conversion within 30/60 days	Monthly by acquisition cohort
Retention	Month-1/3/6 cohort retention; 30/60/90-day inactivity	Monthly
Data quality	Missing Returns %, duplicate rate, valid dates/amounts	Each data refresh

8. Limitations and Conclusion

This is a descriptive segmentation and retention analysis, not proof of causal campaign impact. The dataset does not specify currency, a distinct order identifier, marketing exposure, acquisition source, or a business-defined churn window. The archive contains another similarly named customer dataset; this report uses ecommerce_customer_data_large.csv as the primary source. Confirm this is the intended source before submission or production use.

In conclusion, Alfido Tech can use RFM segments to prioritize retention activity, focus win-back effort on inactive/high-frequency customers, nurture newer buyers toward a second purchase, and protect valuable repeat customers. Controlled experiments and improved data definitions will make these actions more reliable over time.

Prepared as a customer analytics project report.